

Breaking MAG:

From Metagenomics
Pipeline Selection to
Functional Discovery

This thesis represents a comprehensive suite of bioinformatics tools that enable correct pipeline selection, fair workflow benchmarking and reproducible workflow development, complemented with an end-to-end metagenomics study that reveals novel microbial diversity in Colombian rice straw degradation systems and an advanced deep learning strategy to perform protein functional characterization.

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Chapter 1

High-throughput metagenomics enables genome-resolved analyses of complex microbial communities, although analytical outcomes remain highly dependent on pipeline selection, benchmarking strategies, and reproducibility. This chapter is composed of three peer-reviewed published papers encompassing a suite of tools designed to guide researchers through their metagenomics analysis. This suite is composed of 2Pipe¹, an online application that supports the identification of the correct pipeline to assemble metagenome-assembled-genomes (MAGs) according to the user needs, MAGFlow and BigMAG², a dual tool that (pipeline and dashboard) provides a strategy for standardized benchmarking and comparative analysis, and TaxoFlow³, a tutorial that offers a step-by-step framework for building reproducible workflows. Collectively, these tools encourage informed methodological decisions and transparent metagenomic analyses.

Chapter 2

Rice straw (RS) is usually burnt after every harvest season, generating serious environmental and human health concerns. Thus, in this chapter, we performed a genome-resolved metagenomics study of RS degradation in Colombian field experiments. We recovered 146 MAGs of which a majority (67%) represent likely novel taxa (Fig. 1). Moreover, our taxonomic and functional analysis highlighted MAGs with genomic features for lignocellulose biotransformation potential, offering insights into the microbes driving RS decomposition. These outcomes provide a valuable resource for exploring uncharacterized microbial diversity and for future development of bio-based RS valorization strategies. The methodology and the approach followed to reconstruct and annotate the MAGs are available through a peer-reviewed published paper⁴.

FIGURE

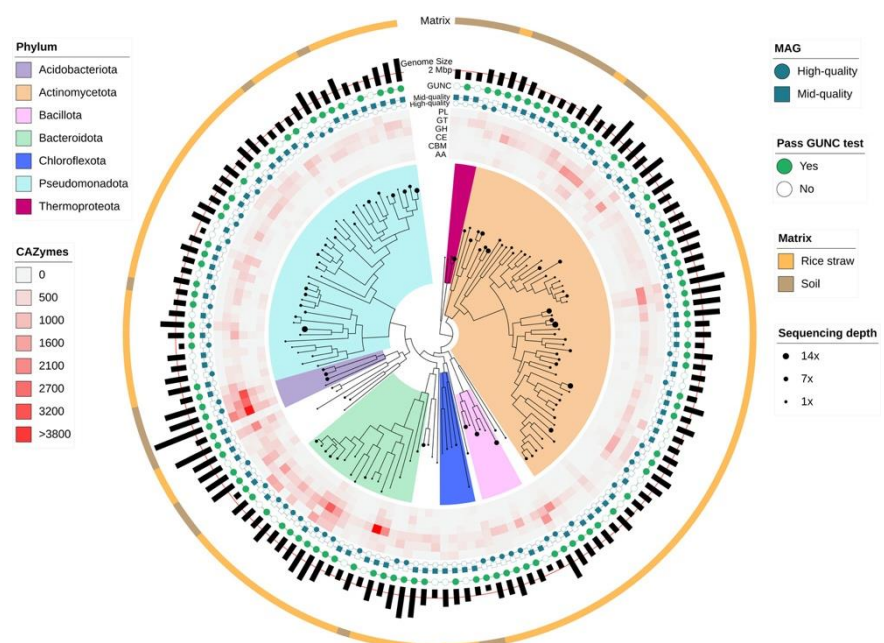


Fig. 1 | Phylogenomic organization of the 146 MAGs generated by GTDB-Tk2. The visualization includes phylogenetic relationship among them (tree), specific highlighted phylum clades (colored sectors), MAG sequencing depth (tips of the branches), Carbon Active EnZyme (CAZyme) class normalized abundance (red scaled heatmap), quality assessment as either high-quality or mid-quality MAG based on checkM2 measurements (blue binary), GUNC test passing (green binary), and genome size in Mb (outer black bars).

My thesis in brief

Chapter 3

Functional annotation remains a major bottleneck in metagenomic analysis, particularly for enzymes involved in complex carbohydrate degradation. In this chapter, we analyzed enzymes involved in lignocellulose decomposition by fine-tuning the model ProteinBERT to boost the detection of carbohydrate-active enzymes (CAZy). Briefly, we exposed the model to a subset of the CAZy database to direct its pre-training towards the discrimination of enzymes related to lignocellulose bio-transformation. The resulting fine-tuned model (CAZyBERT) excels at classifying protein sequences belonging to the subset of the CAZy database. Further, when tested with structural homologs obtained from the ESM Metagenomics Atlas, CAZyBERT seems able to capture nuances related with functional features conferred by protein structure, outperforming tools like CAZyLingua, CAZyFoldDB and dbCAN3 (HMMER and DIAMOND) (Fig. 2).

FIGURE

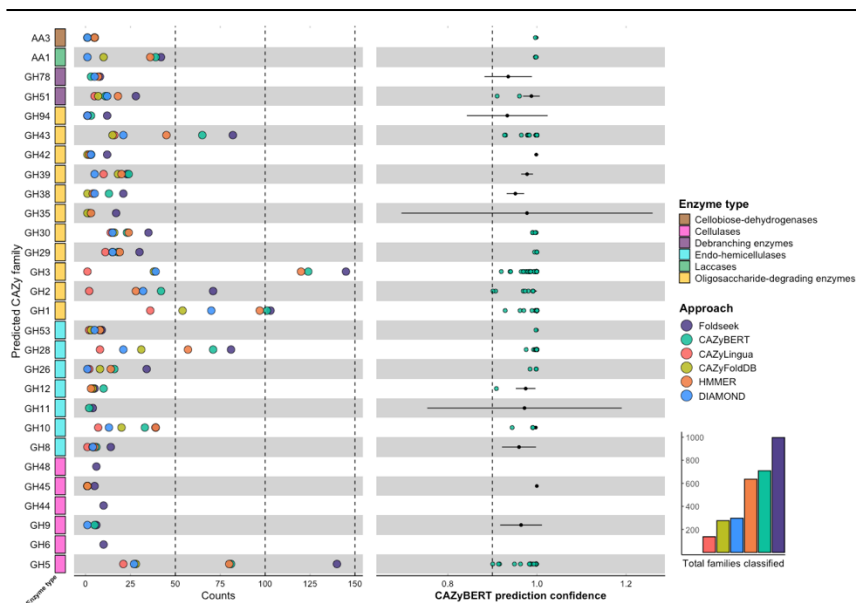


Fig. 2 | Classification of protein sequences by different models. Lignocellulose-degradation CAZy family's annotation by CAZyBERT, CAZyLingua, CAZyFoldDB-annotate (labeled as CAZyFoldFoldDB), and HMMER and DIAMOND. Query sequences were obtained by searching the ESMAtlas-30 using Foldseek, with protein structures retrieved from the AlphaFold DB as queries. The AlphaFold structures are proteins with known CAZy annotations. Colored bars on the left indicate the type of enzyme: cellulases, cellobiose-dehydrogenases, debranching enzymes, endo-hemicellulases, laccases and oligosaccharide-degrading enzymes. On the right panel, it is shown the distribution of prediction confidence scores produced by CAZyBERT; colored points are confidence outlier values. The inset bar plot (bottom right) summarizes the total number of CAZy families detected by each method. The label Foldseek represents the gold standard.

PUBLICATIONS

1. **Yepes-García J.** & Falquet L. (2026). 2Pipe starts with a question: matching you with the correct pipeline for MAG reconstruction. *mSystems*, [0:e00844-25](#).
2. **Yepes-García J.** & Falquet, L. (2024). Metagenome quality metrics and taxonomical annotation visualization through the integration of MAGFlow and BigMAG. *F1000Research*, [13:640](#).
3. **Yepes-García J.** & Falquet L. (2026). TaxoFlow: a step-by-step tutorial to build a Nextflow pipeline for metagenomics taxonomic classification. *Gigabyte*, [187](#).
4. **Yepes-García J.**, Novoa-Montenegro N., Otero-Jiménez V., Uribe-Vélez D., Barreto-Hernández E. & Falquet L. (2025). Genome-Resolved Metagenomics of Rice Straw Degradation Experiments in Colombian Fields. *Scientific Data*, [12:1](#).
5. Falquet L, **Yepes-García J.**, Mateus Gonzalez I. & Barreto-Hernandez E. (2024). Report on the Swiss-Colombian workshop 2024: "Metagenomics data analysis of mixed microbial communities". *EMBNet journal*, [30:e1063](#).
6. Alvarez Saravia D., Rosenbaum A., Straub D., Downie J., Borry M., Fedewa G., Hübner A., Lundin D., **Yepes-García J.**, McDonald J., Nahnsen S., Köhn L., Uribe-Paredes R., Navarete M.A., Warinner C., nf-core community & Fellows Yate J.A. (2026). Community-driven updates for comprehensive long-read metagenomics and enhanced binning in nf-core/mag v5. *Bioinformatics*, [0:0](#).

AWARDS

- Swiss Institute of Bioinformatics (SIB) Remarkable Output during the Basel Computational Biology Conference [BC]² 2025.
- The Best Poster in Bioinformatics and Systems Biology during the Life Sciences Switzerland (LS²) Annual Meeting 2025.

TRAINER

- Introduction to Metagenomics Data Analysis of Microbial Communities.
- Nextflow in Action: Build Smarter, Faster, Reproducible Pipelines.
- Interactive Visualisation with Python.

More information: <https://www.sib.swiss/training/upcoming-training-courses>.

ABOUT ME



I started my doctoral studies under the Swiss Government Excellence Scholarship program after finishing my master's program in engineering at the University of Antioquia. During my time in Fribourg, I have been FGLM Student Committee Member, CUSO StarOmics Student Representative and Nextflow Ambassador.

Supported by:

